

Honors Intro Micro | Homework 6 Demo

Question 1 | Wandmaker's Dilemma

Suppose the demand for wands is given by

$$P = 100 - Q$$

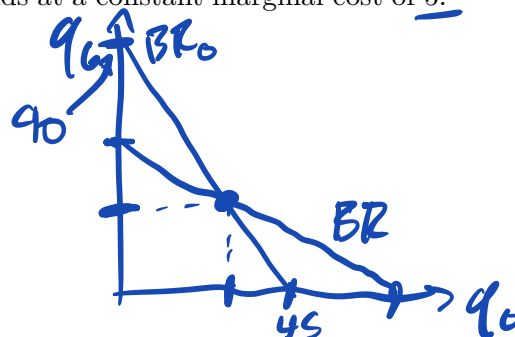
$$q_G + q_O = Q$$

and there are two main wand sellers, Olivander and Gregorovitch. Olivander can make wands at a constant marginal cost of 10, while Gregorovitch can make wands at a constant marginal cost of 5. The marginal revenue for Olivander is

$$MR_O = 100 - 2q_O - q_G$$

and for Gregorovitch it is

$$MR_G = 100 - 2q_G - q_O$$



Units are in galleons and stones. Calculate the Nash equilibrium level of output for the two wandmakers and put it in the blanks below.

Note. This Oligopoly is not symmetric. So the trick we used in class won't work here. Simply solve for both firm's optimization decisions separately.

Part A. Olivander's Quantity

What is the Nash equilibrium quantity for Olivander?

These give us the BR for both firms.

$$MC_O = MR_O$$

$$10 = 100 - 2q_O - q_G$$

$$2q_O = 90 - q_G$$

$$q_O = 45 - \frac{1}{2}q_G$$

$$q_O = 45 - \frac{1}{2}\left(\frac{95}{2} - \frac{1}{2}q_O\right) = 45 - \frac{95}{4} + \frac{1}{4}q_O$$

$$MC_G = MR_G$$

$$5 = 100 - 2q_G - q_O$$

$$2q_G = 95 - q_O$$

$$q_G = \frac{95}{2} - \frac{1}{2}q_O$$

$$\left(1 - \frac{1}{4}\right)q_O = \frac{120}{4} - \frac{95}{4}$$

$$\frac{3}{4}q_O = \frac{25}{4}$$

$$q_O^* = \frac{25}{3}$$

SOLUTION

Part B. Gregorovitch's Quantity

What is the Nash equilibrium quantity for Gregorovitch?

$$q_G = \frac{95}{2} - \frac{1}{2}q_0 = \frac{95}{2} - \frac{1}{2} \cdot \frac{95}{3} = \frac{95}{2} - \frac{95}{6} = \frac{285 - 95}{6}$$

$$q_G = \frac{200}{6} = \boxed{\frac{100}{3} = q_G^*} \quad \text{SOLUTION}$$

Part C. Equilibrium Price

What is the Nash equilibrium price in this market?

$$P = 100 - Q = 100 - \frac{185}{3} = \frac{300 - 185}{3}$$

$$\boxed{P^* = \frac{115}{3}}$$

Part D. ~~Manufacturing~~ ^{Taxing} Wands

In a highly unethical move, the Ministry of Magic imposed a 10 galleon ~~increase~~ ^{tax} on wand sales, ensuring only the wealthiest were able to attend wizarding school. Model this as a 10 galleon increase in the marginal cost of making a wand and find the Nash equilibrium quantity and price after the tax.

$$MC_G = 5 + 10$$

$$MC_0 = 10 + 10$$

$$15 = 100 - 2q_G - q_0$$

$$20 = 100 - 2q_0 - q_G$$

$$2q_G = 85 - q_0$$

$$2q_0 = 80 - q_G$$

$$q_0 = 85 - 2q_G$$

$$2 \cdot 85 - 4q_G = 80 - q_G$$

$$q_0 = 85 - 2q_G$$

$$170 - 80 = 3q_G$$

$$q_0 = 85 - 60$$

$$90 = 3q_G$$

$$\boxed{q_0^* = 15}$$

$$\boxed{q_G^* = 30}$$

$$P^* = 100 - 45$$

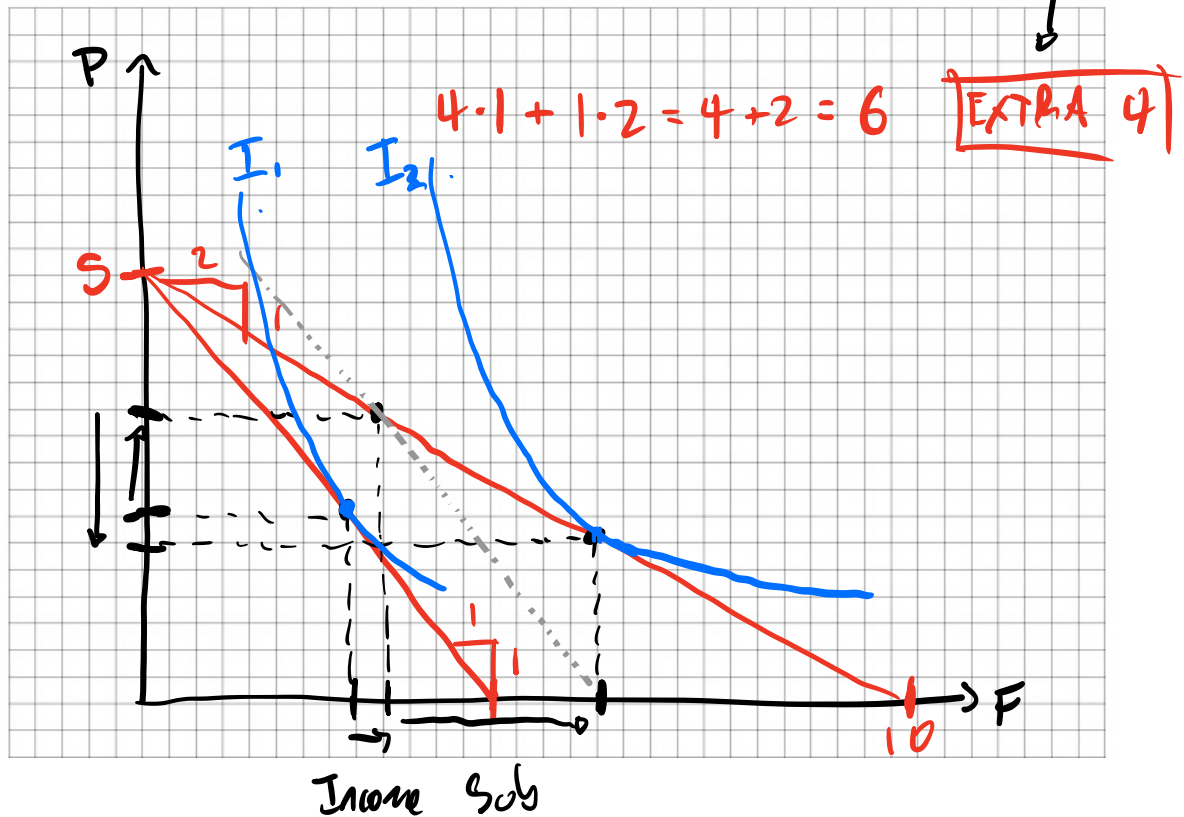
$$\boxed{P^* = 55}$$

Question 2 | Luna's Problem

Luna divides her 10 galleons between pumkin pasties which cost 2 galleons and chocolate frogs which costs 2 galleon (each of which are normal goods). But due to an improved harvesting technique for cocoa, the price of chocolate frogs to gets cut in half right before the winter holiday season.

Part A

Draw Luna's budget constraint before harvesting improved.



Part B

Show the effect of the new harvesting technique on Luna's budget constraint on the graph from Part A.

Part C

Draw Luna's indifference curve on the graph from Part A and show the effect of the new harvesting technology on Luna's optimal consumption bundle assuming that the substitution effect outweighs the income effect for pumkin pasties.

Part D

In response to the price increase around the holiday season, the Ministry of Magic sent every wizard and witch a 10 galleon check in the mail to ensure families are able to afford the treat. Show the effect of the 10 galleons on Luna's optimal consumption bundle.

